Interpolate Grouped Time Series

Description

Interpolates grouped numeric time series to a regular age grid.

Usage

```
interpolate_grouped_time_series(
  data_time_series = NULL,
  grouping_variables = "dataset_name",
  age_variable_name = "age",
  value_variable_name = "value",
  interpolation_method = "linear",
  extrapolation_rule = 1,
  ties_function = base::mean,
  age_min = 0,
  age_max = 12000,
  time_step = 500,
  n_cores = 1,
  show_progress = TRUE
)
```

Arguments

`data_time_series`	A data frame containing grouping, age, and value columns.
`grouping_variables`	A character vector of column names that define independent time series.
`age_variable_name`	Name of the numeric age column.
`value_variable_name`	Name of the numeric value column.

`interpolation_method`	Interpolation method passed to [stats::approx()].
`extrapolation_rule`	Numeric extrapolation rule passed to [stats::approx()].
`ties_function`	Function used by [stats::approx()] to combine tied ages.
`age_min`	Minimum age in the output grid.
`age_max`	Maximum age in the output grid.
`time_step`	Positive spacing between consecutive ages in the output grid.
`n_cores`	Number of workers. Use '1' for sequential 'purrr::map()' execution or a larger integer for parallel 'furrr::future_map()' execution.
`show_progress`	Logical indicating whether supported map operations display progress.

Details

Each unique combination of 'grouping_variables' is interpolated independently with [stats::approx()]. Groups that cannot be interpolated return no rows.

Parallel execution temporarily owns the active 'future' plan. The previous plan is restored when the function exits.

Value

A data frame containing the grouping columns and interpolated age and value columns.

See Also

[stats::approx()]